FIITJEE

ALL INDIA TEST SERIES

FULL TEST – V

JEE (Main)-2021

TEST DATE: 08-02-2021

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 Questions.

  Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

Physics

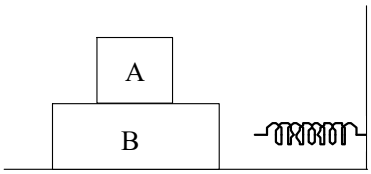
PART – I

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. When someone jumps, one always bends one's knees to lower the centre of mass. A person lowers its centre of mass of the body by 40 cm and then jumps up. One can usually reach a height of 50 cm above your normal height. If a man of mass 80 kg jumps as described above, the work done by the man is (Take g = 10 m/s^2)
   (A) 720J
   (B) 400J
   (C) 320J
   (D) 0

2. A block A is placed over block B having mass m & 2m respectively. Block B is resting on a frictionless surface and there is friction between block A and B. The system of blocks is pushed towards a spring with a velocity v_0 such that A doesn't slip on B by the time the system comes to momentary rest. The correct statement is:



   (A) Work done by friction on A is zero
   (B) Work done by friction on B is $-1/2\text{mv}_0^2$
   (C) Work done by spring on B is $-3/2\text{mv}_0^2$
   (D) Work done by friction on A & B is zero.

3. The magnetic susceptibility of a paramagnetic substance at -73°C is 0.0060, then its value at -173°C will be:
   (A) 0.0030
   (B) 0.0120
   (C) 0.0180
   (D) 0.0045

4. The mean radius of earth is R, its angular speed on its own axis is ω and the acceleration due to gravity at earth's surface is g. What will be the radius of the orbit of a geostationary satellite?
   (A) $\left(R^2g/\omega^2\right)^{1/3}$
   (B) $\left(Rg/\omega^2\right)^{1/3}$
   (C) $\left(R^2\omega^2/g\right)^{1/3}$
   (D) $\left(R^2g/\omega\right)^{1/3}$

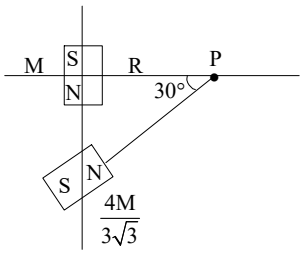
5. Two planets A and B have radius in the ratio $1:6$. A satellite revolving near the surface of planets have same time period of revolution. The ratio of density of planets A and B is:
(A) $1:6$
(B) $6:1$
(C) $1:1$
(D) $36:1$

6. One mole of a diatomic gas undergoes a process $P=\frac{P_0}{\left(1+\frac{V}{V_0}\right)^3}$, where P_0, V_0 are constants. The translational kinetic energy of the gas when $V=V_0$ is given by
(A) $\frac{3P_0V_0}{2}$
(B) $\frac{3P_0V_0}{16}$
(C) $\frac{3P_0V_0}{4}$
(D) $\frac{5P_0V_0}{2}$

7. An experiment measures quantities a, b, c and x the value of x is calculated from $x=ab/c^3$. If the maximum percentage error in a, b and c are 1% , 1% and 2% respectively, then
(A) The error in x is zero
(B) For any particular reading, error in x may be 10%
(C) Maximum percentage error in x is -4%
(D) Maximum percentage error in x is 8%

8. A motorcyclist going around a circular track of radius 50m with a speed of 25 m/s. A static siren outside the circle in the plane of circle is emitting sound of frequency n. How many times (approximately) in an hour will the motor cyclist hear the sound of actual frequency?
(A) 24
(B) 287
(C) 600
(D) 573

9. What will be magnitude of net magnetic field as shown in figure at point P. (Magnetic moment of small magnets are M and $\frac{4M}{3\sqrt{3}}$ respectively) Given that $\left(\frac{\mu_0}{4\pi}\frac{M}{R^3}=B_0\right)$



(A) B_0
(B) $\sqrt{5}B_0$
(C) $\sqrt{3}B_0$
(D) None of these

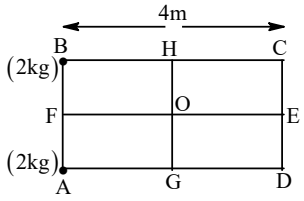
10. The K_α x-ray emission line of tungsten occurs at $\lambda = 0.021\,\text{nm}$. The energy difference between K and L levels in this atom is about:
(A) $0.51\,\text{MeV}$
(B) $1.2\,\text{MeV}$
(C) $59\,\text{KeV}$
(D) $13.6\,\text{eV}$

11. The de-Broglie wavelength of a vehicle moving with velocity v is λ. Its load is changed so that the velocity as well as kinetic energy are doubled. What will be the de-Broglie wavelength now?
(A) λ
(B) 2λ
(C) 4λ
(D) 16λ

12. Light with wavelength $500\,\text{nm}$ falls on a slit of width $10\,\mu\text{m}$ at an angle $30°$ to its normal. Find the angular position of the first minima located on both sides of the central Fraunhofer maximum:-
(A) $33°$ and $27°$
(B) $25°$ and $35°$
(C) $36°$ and $24°$
(D) $28°$ and $32°$

13. Charges are distributed over two spheres of radii R and r such that their surface charge densities are equal. What is the ratio of their potentials?
(A) $\frac{R}{r}$
(B) $\frac{R^2}{r^2}$
(C) $\frac{R^3}{r^3}$
(D) $\frac{R^4}{r^4}$

14. Masses of 2 kg each are placed at the corners B and A of a uniform rectangular plate ABCD as shown in the figure. A mass of 8 kg has to be placed on the plate so that the centre of mass of the system should be at the centre O. Then the mass should be placed at:



(A) $1\,\text{m}$ from O on OE
(B) $2\,\text{m}$ from O on OF
(C) $2\,\text{m}$ from O on OG
(D) $2\,\text{m}$ from O on OH

15. A ball, moving with a speed of 9 m/s, strikes an identical stationary ball such that after the collision, the direction of each ball makes an angle of 30° with the original line of motion. The speeds of the two balls after the collision are, respectively,

(A) $3\sqrt{3}\, m/s,\ 3\, m/s$

(B) $3\, m/s,\ 3\sqrt{3}\, m/s$

(C) $3\, m/s,\ 3\, m/s$

(D) $3\sqrt{3}\, m/s,\ 3\sqrt{3}\, m/s$

16. The distance between an object and a screen is 100cm. A lens produces an image on the screen when placed at either of the two positions 40cm apart. The power of the lens (in D) is nearly:-

(A) 4
(B) 5
(C) 3
(D) 6

17. The electric potential at a point (x, y, z) is given by $V = -x^2 y - xz^3 + 4$. The electric field at that point is:

(A) $\vec{E} = \hat{i}\, 2xy + \hat{j}\left(x^2 + y^2\right) + \hat{k}\left(3xz - y^2\right)$

(B) $\vec{E} = \hat{i}\, z^3 + \hat{j}\, xyz + \hat{k}\, z^2$

(C) $\vec{E} = \hat{i}\left(2xy - z^3\right) + \hat{j}\, xy^2 + \hat{k}\, 3z^2 x$

(D) $\vec{E} = \hat{i}\left(2xy + z^3\right) + \hat{j}\, x^2 + \hat{k}\, 3xz^2$

18. Two sources of sound placed close to each other, are emitting progressive waves given by $y_1 = 4\sin 600\pi t$ and $y_2 = 5\sin 608\pi t$

An observer located near these two sources of sound will hear

(A) 4 beats per second with intensity ratio 25: 16 between waxing and waning.
(B) 8 beats per second with intensity ratio 25: 16 between waxing and waning.
(C) 8 beats per second with intensity ratio 81: 1 between waxing and waning.
(D) 4 beats per second with intensity ratio 81: 1 between waxing and waning.

19. A potentiometer circuit has been set up for finding the internal resistance of a given cell. The main battery, used across the potentiometer wire, has an emf of 2.0 V and a negligible internal resistance. The potentiometer wire itself is 4m long. When the resistance R, connected across the given cell, has values of

(i) Infinity

(ii) 9.5Ω the balancing lengths on the potentiometer wire are found to be 3m and 2.85 m, respectively. The value of internal resistance of the cell is

(A) 0.25Ω
(B) 0.95Ω
(C) 0.5Ω
(D) 0.75Ω

20. The efficiency of Carnot engine is 50% and temperature of sink is 500 K. If temperature of source is kept constant and its efficiency raised to 60%, then the required temperature of sink will be

(A) 100K
(B) 600K
(C) 400K
(D) 500K

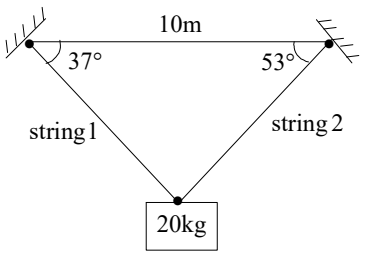
SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

21. A simple telescope consisting of an objective of focal length $60\,\text{cm}$ and a single eye lens of focal length $5\,\text{cm}$ is focussed on a distant object in such a way that parallel rays emerge from the eye lens. If the objects subtend an angle of $2°$ at the objective then find the angular width of the image (in degree).

22. The optical system consists of a thin convex lens of focal length 30cm and a plane mirror 15cm right of the lens. An object is placed 15cm left of the lens. Find the distance (in cm) of final image from lens.

23. A mass of 20kg is hanging with support of two strings of same linear mass density. Now pulses are generated in both strings at same time near the joint at mass. If ratio of time, taken by a pulse travel through string 1 to that taken by pulse on string 2 is $\frac{a}{b\sqrt{c}}$ then fill the value of $a+b+c$.

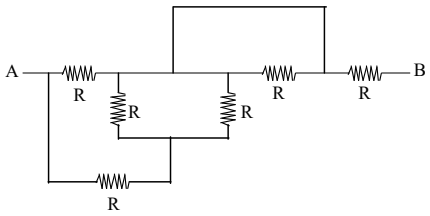


24. The speed of light in glass of refractive index 1.5 is $2\times10^8\,m/s$. In a certain liquid the speed is $2.5\times10^8\,m/s$. The refractive index of the liquid is

25. For a partially polarized light, we can define the degree of polarization. The degree of polarization of light is given by $P=\frac{I_{max}-I_{min}}{I_{max}+I_{min}}$. Here I_{max} and I_{min} are maximum and minimum possible intensities in two mutually perpendicular directions. For a given partially polarized light beam, the value of P is 0.25. The approximate ratio of maximum and minimum amplitudes in two mutually perpendicular directions is:

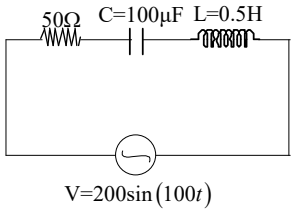
26. The optical powers of the objective and the eyepiece of a microscope are equal to 100 and 20D respectively. The angular magnification is equal to 50 when the eye is relaxed. What will the magnification of the microscope be when the distance between the objective and the eyepiece is increased by 2.0cm but the eye is still relaxed?

27. If the equivalent resistance between A and B is KR. Fill the value of K. (Take $R=1\Omega$)



28. If a dipole $p\hat{i}$ and a point charge placed at $(0, y)$ and at the origin respectively, create no net electric field at point $(x, x+y)$. Find the ratio of $\frac{y}{x}$?

29. In the given LCR AC circuit, the effective current (in ampere) flowing through the circuit is.



30. A string of length 1m fixed at both ends is vibrating in 3rd overtone. Tension in string is 200N and linear mass density is 5 gm/m. Find the frequency of these vibrations in Hz.

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. The normal boiling point and ΔH_{vap} of a liquid 'X' are 400 K and 40kJ mol^{-1} respectively. Assuming ΔH_{vap} to be constant, which of the following is correct?

(I) $\Delta S_{vap} > 100\, JK^{-1}mol^{-1}\ at\ 400\, K$ and $0.5\,\text{atm}$

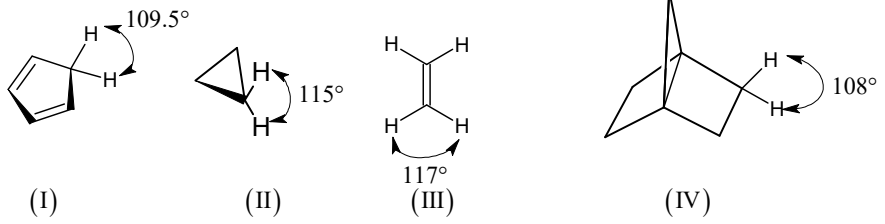
(II) $\Delta S_{vap} < 100\, JK^{-1}mol^{-1}\ at\ 400\, K$ and $1\,\text{atm}$

(III) $\Delta S_{vap} < 100\, JK^{-1}mol^{-1}\ at\ 400\, K$ and $2\,\text{atm}$

(IV) $\Delta S_{vap} = 100K\, JK^{-1}mol^{-1}\ at\ 400\, K$ and $1\,\text{atm}$

(A) II and IV
(B) II only
(C) I and III
(D) I, III and IV

32. Selected bond angles of six hydrocarbons are shown below. Arrange these hydrocarbons according to their pK_a values, from the lowest to the highest.



(I) (II) (III) (IV)



(V) (VI)

(A) V<I<VI<II<III<IV
(B) IV<I<II<III<V<VI
(C) II<IV<I<VI<V<III
(D) I<V<IV<III<II<VI

33. Select the correct set of statement/s:
(I) Work done by the surrounding in case of infinite step expansion is more than single stage expansion
(II) Irreversible work is always greater than reversible work.
(III) On an ideal gas in case of single stage expansion and compression system as well as surrounding are restored back to their original states.
(IV) If gas is in thermodynamic equilibrium is taken from state A to state B, by four successive single stage expansions. Then we can plot maximum 4 points on the P-V indicator diagram.

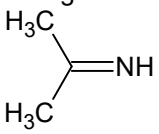
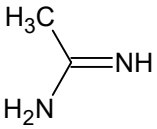
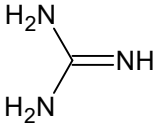
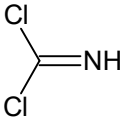
(A) II
(B) I, II, III, IV
(C) II, IV
(D) I, II, IV

34. Three closed vessel A, B and C are at the same temperature and contain gases which obey the Maxwellian distribution of velocities. Vessel A contains only O_2, B only N_2, and C, a mixture of equal quantities of O_2 and N_2. If the average speed of the O_2 molecules in vessel A is V_1, that of the N_2 molecules in vessel B is V_2, the average speed of the O_2 molecules in vessel C is:

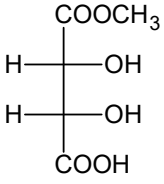
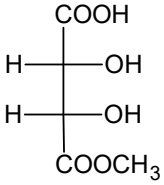
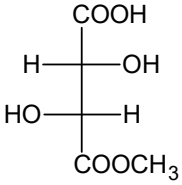
(A) $(V_1+V_2)/2$
(B) V_1
(C) $\sqrt{V_1V_2}$
(D) $\sqrt{3kT/M}$

Where M is the mean molar mass of the mixture

35. Which among the following compounds is most basic?

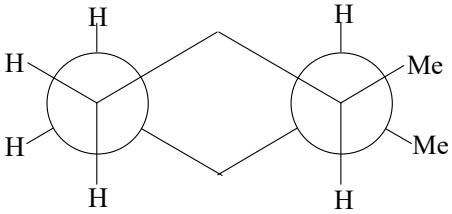
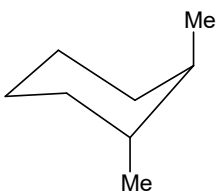
(A) 
(B) 
(C) 
(D) 

36. The correct statement about the compound A, B and C.


(A)

(B)

(C)

(A) 'A' and 'B' are identical
(B) 'A' and 'B' are diastereomers
(C) 'A' and 'C' are enantiomers
(D) 'A' and 'B' are enantiomers

37. The correct stability order of the following species is:


(a)

(b)

(c)

(A) c<a<b
(B) c=b<a
(C) c<a=b
(D) a=b=c

38. Total number of d-orbitals involved in the hybridization of central metal ion in complex ion $\left[MnO_4\right]^-$ is:

(A) 6
(B) 4
(C) 2
(D) 3

39. Which of the following processes helps to react Al with cold water readily to evolve H_2 ?

(A) Amalgamating with Hg
(B) Galvanizing with Zn
(C) Anodizing with dil. H_2SO_4
(D) None of these

40. The lowest melting solid among the following elements is:

(A) Si
(B) Pb
(C) Ge
(D) Sn

41. If the oxidation state of central atom in BH_3 is $+3$, then the oxidation states of the central atom in SiH_4 and CH_4 are, respectively.

(A) $+4, -4$
(B) $+4, +4$
(C) $-4, +4$
(D) $-4, -4$

42. Choose the correct lattice energy order:

(A) LiH<NaH<KH
(B) $MgH_2 > CaH_2 < SrH_2$
(C) LiH<NaH>KH
(D) $MgH_2 > CaH_2 > SrH_2$

43. Read the following statements:

(I) Al has greater affinity than Fe, for oxygen.
(II) Cast iron has impurity of zinc and lead.
(III) Refining of nickel is done by vapour phase refining.
(IV) In cyanide process, oxygen and zinc dust are used as oxidizing agent and reducing agent, respectively.

Choose the all correct set of statements.

(A) I,III
(B) II,III,IV
(C) I,III,IV
(D) I,IV

44. Which of the following complexes does not show optical activity?

(A) $\left[\mathrm{Co(EDTA)}\right]^{-}$

(B) $\left[\mathrm{Pt(bn)_2}\right]^{2+}$

(C) $\left[\mathrm{Pt(pn)_2}\right]^{2+}$

(D) $\left[\mathrm{Pt(en)_2}\right]^{2+}$

45. Which of the following compounds produce acid having basicity of two on hydrolysis?

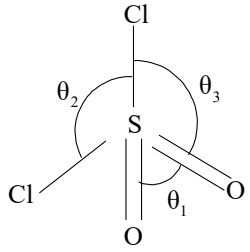
(A) $\mathrm{NCl_3}$

(B) $\mathrm{Al_2(CH_3)_6}$

(C) $\mathrm{SO_2Cl_2}$

(D) $\mathrm{Mg_3N_2}$

46. Which of the following statements is correct for



(A) It contains $\mathrm{p}_{\pi}-p_{\pi}$ and $\mathrm{p}_{\pi}-d_{\pi}$

(B) It has regular tetrahedral geometry.

(C) $\theta_1 > \theta_3$

(D) Plane which contains maximum number of atom is 4.

47. Match the atoms/ions with their corresponding electron affinity. (eV/atom).

Column – I		Column – II	
(P)	F	(1)	3.4
(Q)	F^+	(2)	17.4
(R)	Cl	(3)	13
(S)	Cl^+	(4)	3.6

Code:

	P	Q	R	S
(A)	1	3	4	2
(B)	2	1	3	4
(C)	1	2	4	3
(D)	3	1	2	4

48. Potassium selenate $\left(\mathrm{K_2SeO_4}\right)$ is isomorphous with potassium sulphate $\left(\mathrm{K_2SO_4}\right)$. If potassium selenate contains 50% Se, the atomic mass of Se is:

(A) 122

(B) 142

(C) 162

(D) 112

49. Photo-electric emission is observed from a surface for frequencies ν_1 and ν_2 of the incident radiation $(\nu_1 > \nu_2)$. If the maximum kinetic energies of the photoelectrons in the two cases are in the ratio $1:k$ then the threshold frequency v_0 is given by:

(A) $\dfrac{v_2 - v_1}{k-1}$

(B) $\dfrac{kv_1 - v_2}{k-1}$

(C) $\dfrac{kv_2 - v_1}{k-1}$

(D) $\dfrac{v_2 - v_1}{k}$

50. A soap of sodium stearate in solution starts micellisation at $1.2\times10^{-3} M$. A colloidal (micelle) particle, on an average contains 2.4×10^{13} molecules of soap in $1\,\text{mm}^3$. The average number of anions (stearate ions) is:

(A) 120

(B) 90

(C) 60

(D) 30

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

51. From the following data. Calculate the magnitude of standard enthalpy of formation of propane (in KCal) $\Delta_f H^- CH_4 = -17\,\text{kcal mol}^{-1}$.

$\Delta_f H^- C_2H_6 = -24\,\text{kcal mol}^{-1}; BE(C-H) = 99\,\text{kcal mol}^{-1}\ (C-C) = 84\,\text{kcal mol}^{-1}$

52. At $25°\text{C}$ and 1 atm pressure, the partial pressure in equilibrium mixture of N_2O_4 and NO_2, are 0.7 and 0.3 atm, respectively. Calculate the partial pressure of these gasses when they are in equilibrium at $25°\text{C}$ and a total pressure of 10 atm.

53. 20 g of hydrogen and 128 g of oxygen are contained in a 20L flask at $200°\text{C}$. Calculate the total pressure of the mixture. If a spark ignites the mixture, what will be the final pressure?

54. A stationary He^+ ion emits a photon corresponding to the first line (H_α) of the Lyman series. The photon thus emitted strikes a H atom in the ground state. Find the velocity of the photoelectron ejected out of the hydrogen atom, if its $y\times10^6 m/s$ then the value of y is?

55. The unbalanced equation for the reaction of P_4S_3 with nitirate in aqueous acidic medium is give below.

$P_4S_3 + NO_3^- \rightarrow H_3PO_4 + SO_4^{2-} + NO$

The number of mol of water required per mol of P_4S_3

56. The solubility of a salt of weak acid (AB) at pH 3 is $Y \times 10^{-3} mol\, L^{-1}$. The value of Y is___ (Given that the value of solubility product of AB $(K_{sp}) = 2 \times 10^{-10}$ and the value of ionisation constant of $HB(K_a) = 1 \times 10^{-8}$)

57. The equivalent conductivity of CH_3COONa, HCl and $NaCl$ at infinite dilution are 91.6, 425.0 and $128.1 S cm^2 eq^{-1}$ respectively. If degree of dissociation of $0.1N$ acetic acid is 0.001, then the equivalent conductivity at this concentration of acetic acid in $S cm^2 eq^{-1}$ is $x \times 10^{-2}$, then the value of x is?

58. Find out the percentage of the reactant molecules crossing over the energy barrier at $70K$, given that $\Delta H_{70} = 0.12\, kcal, E_{a(b)} = +0.02\, kcal$. [Take $e = 2.718$]

59. A decinormal solution of potassium ferrocyanide is 50% dissociated at $300K$. If osmotic pressure of the solution is $A \times 10^5\ Nm^{-2}$ then the value of A is.

(Given $R = 8.314\, JK^{-1}\, mol^{-1}$)

60. $60\,mL$ of H_2 and $42\,mL$ of I_2 are heated in closed vessel. At equilibrium the vessel contains $28\,mL$ HI. Calculate degree of dissociation of HI. All measurements are made at same P and T.

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. For real valued functions $f_i(x) = \sin 2p_i x$, $\forall$ i = 1, 2, 3; $p_i \in N$ such that period of $f_1(x) - f_2(x) + 3f_3(x)$ is π, period of $f_1(x) + 2f_2(x)$ is $\frac{\pi}{2}$ and period of $f_1(x) + 10f_3(x)$ is $\frac{\pi}{3}$ then $\min(p_1 + p_2 + p_3)$ is
(A) 5
(B) 10
(C) 11
(D) 8

62. If ABCD is a tetrahedron such that each ΔABC, ΔABD and ΔACD has a right angle at A. If $\text{ar}(\Delta ABC) = k_1$, $\text{ar}(\Delta ABD) = k_2$, $\text{ar}(\Delta BCD) = k_3$ then $\text{ar}(\Delta ACD)$ is
(A) $\sqrt{k_1^2 + k_2^2 + k_3^2}$
(B) $\sqrt{\frac{k_1 k_2 k_3}{k_1 + k_2 + k_3}}$
(C) $\sqrt{\left|k_1^2 + k_2^2 - k_3^2\right|}$
(D) $\sqrt{\left|k_2^2 - k_1^2 - k_3^2\right|}$

63. Let $D_r = \{0.a_1a_2a_3\ldots a_r;\ a_i = 0 \text{ or } 1,\ 1 \le i \le r-1,\ a_r = 1\}$ be a set of decimal fractions, let T_r be the number of distinct elements in set D_r and S_r be their sum then $\lim\limits_{r\to\infty} \frac{S_r}{T_r}$ is
(A) $\frac{2}{5}$
(B) $\frac{7}{19}$
(C) $\frac{1}{18}$
(D) $\frac{1}{31}$

64. In a ΔMAP, on sides MA and AP squares are drawn. If P and D are on the same side of AM and M, E lies on opposite side of AP where D and E are the centres of the squares on MA and AP respectively then angle between MP and DE is
(A) $\frac{\pi}{2}$
(B) $\frac{\pi}{4}$
(C) $\frac{\pi}{3}$
(D) $\frac{\pi}{6}$

65. A boy have 32 cards out of which 10 cards, each of blue, green and red colours, have denominations as $\{2^1, 2^2, \ldots, 2^{10}\}$ and one black and one white each has value unity. The number of ways in which the boy can get a sum of 2012 if he can choose any number of cards, is
(A) $(1001)^2$
(B) $(1002)^2$
(C) $(1007)^2$
(D) $(1111)^2$

66. If $f(x) = [2x] \sin 3\pi x$ and $f'(k^+) = \lambda k\pi(-1)^k$, (where [.] denotes the greatest integer function and $k \in N$) then the value of λ is
(A) 1
(B) 3
(C) 5
(D) 6

67. The circum circle of ΔABC is $x^2 + y^2 - 5x - 4y + 6 = 0$, if a parabola $(k + 1)y^2 = x$ have sides AB, BC, CA as tangents then the value of k is/are:
(A) $-\frac{7}{8}$
(B) $-\frac{11}{13}$
(C) $\frac{2}{3}$
(D) $\frac{4}{7}$

68. If $f(x) = |x - 1| + |x - 4| + |x - 9| + \ldots + |x - 2500|$ $\forall$ $x \in R$, where m and n are respectively the number of integral points at which f(x) is non-differentiable and f(x) has absolute minimum, then (m, n) is
(A) (50, 56)
(B) (50, 54)
(C) (49, 55)
(D) None of these

69. Consider the matrices $A = \begin{bmatrix} 2 & 1 \\ 4 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix}$ and $C = \begin{bmatrix} 3 & -4 \\ -2 & 3 \end{bmatrix}$, then the value of $\text{tr}(A) + \text{tr}\left(\frac{ABC}{2}\right) + \text{tr}\left(\frac{A(BC)^2}{4}\right) + \ldots\infty$ is
(A) 2
(B) $\frac{1}{2}$
(C) 4
(D) 6

70. The probabilities of solving a problem correctly by A and B are $\frac{1}{8}$ and $\frac{1}{12}$ respectively. Given that they obtain the same answer after solving a problem and the probability of a common mistake by them is $\frac{1}{1001}$, then probability that their solution is correct is
(Assuming that if they commit different mistake then their answers will differ)

(A) $\frac{77}{96}$

(B) $\frac{14}{15}$

(C) $\frac{2}{5}$

(D) $\frac{13}{14}$

71. The region M consists of all the points (x, y) satisfying the inequalities $y \geq 0$, $y \leq x$ and $y \leq 2 - x$. The region N which varies with the parameter t consists of all points which satisfy $t \leq x \leq t + 1 \ \forall \ t \in [0, 1]$, then min(ar (M $\cap$ N)) is

(A) $\frac{1}{2}$ sq. units

(B) $\frac{3}{4}$ sq. units

(C) $\frac{7}{8}$ sq. units

(D) $\frac{16}{3}$ sq. units

72. The domain of the function f(x) = $\sqrt{\frac{(\sin x - e^x)(\cos x - 3)(\ln x + 2)}{\left(x^4 - x + \frac{1}{2}\right)(2x-7)^{30}}}$ belongs to

(A) $\left[\frac{1}{e^2}, \frac{7}{2}\right) \cup \left(\frac{7}{2}, \infty\right)$

(B) $\left(-\frac{1}{2}, 0\right)$

(C) $\left(0, \frac{1}{e^5}\right)$

(D) ϕ

73. Let (a_i, a_{i+1}), for $i = 1, 2, 3, ...n$, be different points lying on the fixed line $y = mx$ in the first quadrant. If A, G, and H are respectively the arithmetic, geometric and harmonic means of abscissae of these points then A, G, H are in:

(A) Arithmetic progression
(B) Geometric progression
(C) Harmonic progression
(D) None of these

74. If $P = \int_{-1}^{2}\left(\frac{\sqrt{x^2+1}+x-1}{\sqrt{x^2+1}+x+1}\right)dx$ and $Q = \int_{1}^{2}\left(\frac{x}{\sqrt{x^2+1}+1}\right)dx$ then $\frac{P}{Q}$ is equal to

(A) $\ln\frac{2}{5}$

(B) $\frac{2}{3}$

(C) $\frac{1}{2}$

(D) 1

75. If α, $\beta(\alpha < \beta)$ are the roots of the equation $g(x) = ax^2 + bx + c = 0$ and $f(x)$ be an even function, then $\int_{\alpha}^{\beta}\frac{e^{f\left(\frac{g(x)}{x-\alpha}\right)}}{e^{f\left(\frac{g(x)}{x-\alpha}\right)}+e^{f\left(\frac{g(x)}{x-\beta}\right)}}dx$ is equal to

(A) $\frac{1}{2}(\beta-\alpha)$

(B) $\frac{1}{2}(\beta^2-\alpha^2)$

(C) $\frac{1}{2}(\beta+\alpha)$

(D) 0

76. A random variable X has the following probability distribution values of X

X:	0	1	2	3	4	5	6	7
P(X):	0	K	2k	2k	3k	k^2	$2k^2$	$7k^2 + k$

then which is true

(A) $\frac{k+1}{P(x<6)} = \frac{10}{81}$

(B) $\frac{P(0<x<5)}{P(x\geq 6)} = \frac{81}{19}$

(C) $\frac{k-1}{P(x\geq 6)} = \frac{2}{23}$

(D) $\frac{P(0<x<4)}{P(x<6)} = \frac{4}{493}$

77. If $f:R \to R$ such that $f(x) = \frac{5}{lx^3+x^2+x+m}$ then (l, m) can be

(A) (1, 2)

(B) (0, 1)

(C) (0, –3)

(D) $\left(-3, \frac{3}{2}\right)$

78. Let A = {1, 2, 3}, then the relation R = {(1, 1), (1, 2), (2, 1)} on A is
(A) Reflexive
(B) Transitive
(C) Symmetric
(D) None of these

79. Let $p \& q$ be two statements then $(\sim p \vee q) \wedge (\sim p \wedge \sim q)$ is a
(A) Tautology
(B) Neither tautology nor contradiction
(C) Contradiction
(D) Either tautology or contradiction

80. A vertical lamp – post of height 9 metres stands at the corner of a rectangular field. The angle of elevation of its top form the farthest corner is $30°$ while from another corner it is $45°$ then area of field is:
(A) $81\sqrt{2}\,\mathrm{m}^2$
(B) $9\sqrt{2}\,\mathrm{m}^2$
(C) $81\sqrt{3}\,\mathrm{m}^2$
(D) $9\sqrt{3}\,\mathrm{m}^2$

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. Let there are 4 boys & 4 girls are arranged in a row such that
$m \rightarrow$ Number of ways in which all girls are consecutive
$n \rightarrow$ Number of ways in which exactly 3 boys are consecutive
then the value of $\sqrt{m+n}$ is

82. If the area of the polygon whose vertices are the solutions (in the complex plane) of the equation $x^7+x^6+x^5+x^4+x^3+x^2+x+1=0$ can be exposed in simplest from as $\frac{a\sqrt{b}+c}{d}$ where a,b,c,d are coprime integers then the value of $\frac{a^2+b^2}{c^2+d^2}$ is_____

83. Let $m(x)=\frac{x^{29}}{3}-5x^{12}+3 \& n(x)=29x^{28}-x^3-1$ then the value of $\lim\limits_{x\to\infty} n(x)\int\limits_0^x e^{m(t)-m(x)}dt$ is_____

84. The value of $\sum\limits_{n=0}^{\infty}\left(\frac{1}{3n+1}-\frac{1}{3n+2}\right)$ is equal to_____
[Use $\pi=3.1415 \& \sqrt{3}=1.732$]

85. If the curve $y = x^6 + ax^5 + bx^4 + cx^3 + dx^2 + ex + f$ touches the line $y = mx + n$ at $x = 1, 2, 3$ then the area bounded by these graphs is_____

86. If the combined mean of two groups is $\frac{40}{3}$ & if the mean of one group with 10 observations is 15 then the mean of the other group with 8 observations is equal to_____

87. Let $\vec{a} = 5\vec{p} + 2\vec{q}$ & $\vec{b} = \vec{p} - 3\vec{q}$ be the adjacent side of a parallelogram. If $\left|\vec{p}\right| = 2\sqrt{2}$ & $\left|\vec{q}\right| = 3$ & angle between $\vec{p}$ & $\vec{q}$ is $\frac{\pi}{2}$ then the length of larger diagonal of the parallelogram is_____

88. The value of $\lim\limits_{n\to\infty}\left(\frac{^{3n}C_n}{^{2n}C_n}\right)^{1/n}$ where $n \in N$ is_____

89. The area enclosed by $\left[\frac{|3x+4y|}{5}\right] + \left[\frac{|4x-3y|}{5}\right] = 3$ where $[.] \to$ GIF is_____

90. If $\sin x = 3\sin y$ & $2\cos x = \cos y$ then the value of $\left(\frac{\sin 2x}{\sin 2y} - \frac{\cos 2x}{\cos 2y}\right)$ is_____